

CELLCHECK · REQUIREMENTS

CellCheck requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/03-requirements.md @ 067b48b		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

CellCheck requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs; they will be checked by calculation at TRL 3 and revised with the first trial partner. Status in Table 2 is judged against the estimates in CCK-PRC-001. Two requirements are not met at TRL 2 (R9 and R14) and three are at risk (R3, R10 and R13).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Test the common salvaged cell sizes	18650 and 21700 cylindrical lithium-ion cells (LCO, NMC, NCA), one cell per channel, 8 channels or more	Design review of holders and profiles
R2	Measure capacity accurately	$\pm 2\%$ of reading at a constant 1.0 A discharge from 4.20 V to 2.80 V	Error budget; later comparison with a calibrated reference
R3	Measure cell voltage accurately	± 10 mV from 2.0 to 4.3 V	Data sheet error budget; later comparison with a calibrated meter
R4	Measure DC internal resistance repeatably	Resolution 1 m Ω or better; repeatability ± 3 m Ω or $\pm 5\%$ of reading, whichever is larger, over 5 reinsertions of one cell	Error budget; later repeat test
R5	Grade enough cells	12 cells or more per attended 10 h day on typical 1.8 Ah cells	Cycle-time calculation
R6	Detect self-discharge	Open-circuit voltage change over a 14-day rest measured to ± 3 mV; cells dropping more than 50 mV (default) flagged	Calculation; later repeat readings
R7	Monitor cell temperature	Each cell measured at least once per second to ± 2 K; channel stops above 45 °C or at a rise of more than 8 K above the bench	Design review; later bench test
R8	Keep the unit cool at full load	Load MOSFET cases at 75 °C or less with all channels discharging at 1 A in a 35 °C room	Thermal calculation
R9	Stay safe after a single fault	No single failure (controller crash, stuck-on load MOSFET, open thermistor, failed charger) lets a cell be charged above 4.25 V, heated past 60 °C, or discharged below 2.5 V	Failure mode and effects analysis
R10	Contain a single cell failure	Tray, liner and guard hold ejecta and flame from one venting 21700 cell without spread beyond the tray	Analysis at TRL 3; physical test only at TRL 4 or later
R11	Fit on a bench	500 x 320 x 120 mm or smaller, 5 kg or less without the adapter	Massing model; later weighing
R12	Stay within the concept budget	\$160 or less in parts at quantity one, excluding cells	Priced BOM
R13	Run from a safe supply	Certified 12 V DC adapter; nothing inside the unit above 13 V; no user mains wiring	Design review
R14	Operate without supervision	Grading can run overnight without a person present	Safety review by Amish
R15	Grade and match automatically	Grades per a configurable rule table; groups for any SxP pack with each parallel group's capacity within $\pm 1\%$ of the mean	Run on sample data
R16	Keep a record for every cell	CSV row per cell with ID, source, measurements, grade, group and dates; export over USB or local Wi-Fi with no cloud account	Design review of the file format
R17	Be buildable and open	Off-the-shelf modules, no fine-pitch assembly; hardware CERN-OHL-S-2.0, software MIT	Design review of the parts list

Table 2. Status at TRL 2 (estimates).

ID	Status	Basis
R1	Met by design	Eight holders for 18650 and 21700
R2	Met on estimate	About ± 1 % after one-time calibration, about ± 2 % without
R3	At risk	About ± 10 mV worst case from typical data sheet figures before calibration; unchecked
R4	Met on estimate, unverified	About 0.8 m Ω per count; repeatability depends on contact force and cell temperature
R5	Met, thin margin	About 12 cells per attended 10 h day
R6	Met on estimate	1.25 mV resolution; room temperature changes of a few kelvin shift readings by about 1 mV (assumption)
R7	Met by design	NTC per cell, firmware limits; response time unverified
R8	Met on estimate	Needs 1.1 K/W; a fan-cooled finned bar is expected to give about 0.5 to 0.8 K/W
R9	Not met	Charger modules limit voltage independently, but a stuck-on load MOSFET can drain a cell below 2.5 V and temperature limits rely on firmware. A hardware watchdog and a per-channel undervoltage comparator are proposed, awaiting Amish
R10	At risk	Containment concept only; no analysis yet
R11	Met	About 460 x 300 x 90 mm, about 3.5 kg
R12	Met, about \$1 margin	About \$159 in parts
R13	At risk	Met by design only if the adapter is a certified unit; low-cost adapters without certification are common
R14	Not met	The concept requires a person present while cells charge or discharge; whether and how overnight runs are allowed is a safety trade-off awaiting Amish
R15	Met by design	Serpentine sort within a grade; tolerance to be confirmed on real cells
R16	Met by design	CSV on microSD and local web page
R17	Met by design	Module-based build

■ Assumptions

- A typical salvaged 18650 holds about 1.8 Ah as found and arrives at about 30 % charge.
- A 14-day rest at 4.10 V separates cells with a harmful self-discharge from healthy ones; the 50 mV threshold is a starting value to be tuned with real cells.
- Operators remove cells from packs by hand before grading; disassembly is outside the scope of this design.